NAVEEN VAYILAPALLI

Data Engineer | 3.6 YOE | PySpark • Airflow • Databricks • AWS • Delta Lake

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Professional Summary

Data Engineer with 3.6 years of experience designing, building, and optimizing scalable batch and near-real-time data platforms using PySpark, Spark SQL, and Databricks. Specialized in developing enterprise ETL/ELT pipelines, implementing Delta Lake Medallion Architecture (Bronze–Silver–Gold), and orchestrating workflows via Apache Airflow. Proven expertise in processing 10M+ records daily, performance tuning Spark workloads, and delivering analytics-ready datasets for BI and reporting. Experienced in AWS cloud deployments and Terraform-driven infrastructure automation for production-grade data platforms.

Technical Skills

Big Data: PySpark, Spark SQL, DataFrame API, dbt | **Databases**: PostgreSQL, MySQL, MongoDB, Delta Tables
Data Engineering: ETL/ELT Pipelines, Incremental Loads, Change Data Capture (CDC), Data Quality, Data Transformation
Streaming: Structured Streaming, Watermarking, Window Aggregations
Data Lakehouse: Delta Lake, Medallion Architecture, ACID Transactions, Time Travel, Schema Evolution
Orchestration: Apache Airflow, DAGs, Sensors, XComs, TaskFlow API, SLAs
Data Warehousing: Star Schema, Snowflake Schema, SCD Type 1 & 2, Dimensional Modelling, Data Modelling
Cloud & Platform: AWS Glue, S3, EC2, IAM, VPC, Databricks Unity Catalog and Workflows, Azure Data Factory, ADLS Gen 2
Infrastructure: Terraform (Infrastructure as Code), Environment Provisioning
Programming: Python, Pandas, REST API Integration, Bash/Shell Scripting
Optimization: Partitioning, Broadcast Joins, AQE, Skew Handling, Caching, Shuffle Optimization
BI & Analytics: Power BI, DAX, KPIs, Row-Level Security (RLS), MS Excel
Version Control & Collaboration: Git, GitHub, CI/CD Basics

Professional Experience

Tata Consultancy Services — Data Engineer / PySpark Developer — Mar 2022 – Aug 2025

Designed and delivered end-to-end data engineering solutions across Databricks Lakehouse and Azure Data Platform, building scalable pipelines and enabling enterprise analytics.

- Engineered scalable ETL/ELT pipelines using PySpark, Spark SQL, and DataFrame API, processing 10M+ records daily from SAP MM, flat files, and relational databases (PostgreSQL, MySQL) into Delta Lake on Databricks and ADLS Gen2. Built and orchestrated pipelines using Azure Data Factory (ADF) and Databricks Workflows, enabling automated, reliable data ingestion and transformation across batch workflows.
- Developed incremental ingestion and Change Data Capture (CDC) pipelines, leveraging ADF pipelines + Databricks notebooks to eliminate full loads and ensure near real-time data availability. Implemented Medallion Architecture (Bronze–Silver–Gold) using Delta Tables stored on ADLS Gen2, transforming raw data into curated, analytics-ready datasets with strong data quality checks using dbt.
- Processed large-scale distributed workloads up to 500GB per execution, leveraging Delta Lake features (ACID transactions, schema evolution, time travel) for reliable and auditable data processing. Optimized Spark workloads using partitioning, broadcast joins, caching, shuffle optimization, and Adaptive Query Execution (AQE), reducing pipeline runtime by 40% and improving query performance by 55%.
- Diagnosed and resolved data skew, join inefficiencies, and performance bottlenecks in distributed Databricks environments. Designed and implemented dimensional data models including Star Schema, Snowflake Schema, and SCD Type 1 & Type 2, supporting supply chain and demand forecasting analytics.
- Developed and maintained 20+ production-grade Apache Airflow DAGs using Sensors, XComs, TaskFlow API, retries, and SLA monitoring; improved pipeline reliability and reduced operational delays by 60%. Integrated ADF with Airflow and Databricks for hybrid orchestration across cloud platforms.

- Built real-time data pipelines using Structured Streaming, implementing watermarking and window aggregations for near real-time analytics use cases. Wrote and optimized complex queries using Spark SQL and PostgreSQL, leveraging CTEs, window functions, and advanced query tuning techniques.
- Managed data governance and access control using Databricks Unity Catalog and Azure AD-based access control for ADLS Gen2, ensuring secure and compliant data access. Automated infrastructure provisioning using Terraform (IaC) across AWS and Azure environments, including S3, EC2, IAM, VPC, ADLS Gen2, reducing environment setup time from 3 hours to 15 minutes.
- Developed Power BI dashboards with DAX, KPIs, drill-through analysis, and Row-Level Security (RLS), enabling insights for 150+ business stakeholders. Utilized Git/GitHub and CI/CD pipelines for version control, automated deployments, and release management across ADF, Databricks, and Airflow workflows.

Certifications

- Deep Learning Specialization — Coursera
- IBM Data Science Professional Certificate - Coursera
- Machine Learning Specialization – Coursera

Projects

Real-Time Streaming Data Pipeline

Designed real-time ingestion framework using Kafka and Spark.

- Implemented watermarking for late data handling.
- Managed checkpointing and state storage for fault-tolerant processing.
- Built window aggregations for KPI reporting.
- Stored curated data in Delta Lake.

Tech Stack: Kafka, PySpark, Structured Streaming, Delta Lake, Watermarking

PySpark Airflow ETL Pipeline

- Built production-grade ETL pipelines using PySpark and Delta Lake.
- Implemented Bronze–Silver–Gold Medallion Architecture with data quality checks.
- Developed YAML-driven configurable ingestion frameworks.
- Orchestrated Spark jobs via Airflow TaskFlow API with SLA monitoring.
- Optimized pipelines using partitioning and incremental processing.

Tech Stack: PySpark, Airflow, Delta Lake, YAML
GitHub: github.com/NaveenVayilapalli007/pyspark-airflow-etl-pipeline

Uber Trips Analytics Dashboard

Built end-to-end analytics dashboards tracking trip volume, revenue, and driver KPIs.

- Designed dimensional data models for scalable reporting.
- Developed drill-through and time intelligence reports using DAX.
- Created paginated PDF reports via Power BI Report Builder.
- Automated refresh and distribution using Power BI Service.

Tech Stack: Power BI, SQL, DAX

Education

B.Tech — Computer Science — Rajiv Gandhi University of Knowledge Technologies 06/2015 - 08/2021
CGPA: 9.0 / 10 (85.5%)